

Cross-Mission expiMap Analysis Recovers Established Tissue Responses and Identifies Complementary Pathway Shifts in Mouse Spaceflight Transcriptomes

Authors and affiliations

Jason Trinh^{1,2} (presenting author); **James A. Casaletto**³; **Walter Alvarado**⁴

¹ Space Life Sciences Training Program, NASA Ames Research Center, Moffett Field, California, USA

² University of California, Berkeley, Berkeley, California, USA

³ Blue Marble Space Institute of Science, NASA Ames Research Center, Mountain View, California, USA

⁴ NASA Ames Research Center, Moffett Field, California, USA

Abstract

Mission differences can obscure recurring organ responses to spaceflight. We asked whether expiMap, a pathway-constrained variational autoencoder, could identify recurring biological programs in mouse spaceflight transcriptomes. NASA Open Science Data Repository bulk RNA-sequencing samples were mapped into tissue-matched expiMap models trained on non-spaceflight ARCHS4 data, with approximately 2,000 highly variable genes linked to current mouse Reactome pathways. Each accession or mission project received equal weight in the summary of flight-ground pathway shifts, which were evaluated using conventional gene-set enrichment, held-out-project prediction, three complete reference-query trainings, cell-composition sensitivity analyses, and member-gene review. Four tissues showed recurring, internally supported patterns. In skin, flight was associated with lower chromatin regulation, DNA repair, Hedgehog signaling, sphingolipid metabolism, and cell-junction programs, consistent with reduced tissue maintenance and barrier coordination. Thymus showed lower DNA-repair and cytoskeletal programs, together with a model-specific lower lymphoid-stromal interaction score. Liver showed lower MHC class II antigen-presentation and T-cell receptor scores despite heterogeneous metabolic responses. Spleen showed coordinated lower T-cell receptor signaling, neutrophil degranulation, and C-type lectin receptor signaling across five unconfounded projects and three trainings; all three programs had preranked GSEA false discovery rates below 0.05. The results agree with established tissue responses and motivate tests of skin and thymic tissue maintenance, hepatic immune communication, and splenic innate effector programs. Independent cell-resolved and functional studies are needed to identify which cells drive these pathway shifts and whether they affect biological function.

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